

# Citadel

## The Knowledge Compilation Layer for Enterprise AI

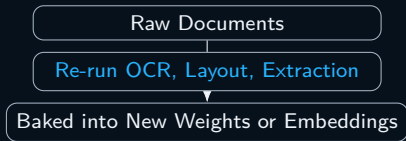
Masum Billal

August 2, 2026

## Every AI System Re-Pays the Same Hidden Cost

Every AI system couples enterprise data to one model's representation — its weights, or its embedding space. Both approaches skip the same intermediary step, and both pay for it the same way, on every model swap.

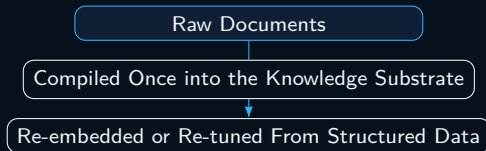
### Without a Knowledge Layer



Full reingestion cost, paid again on every model swap.

Models iterate weekly. Applications pivot monthly.

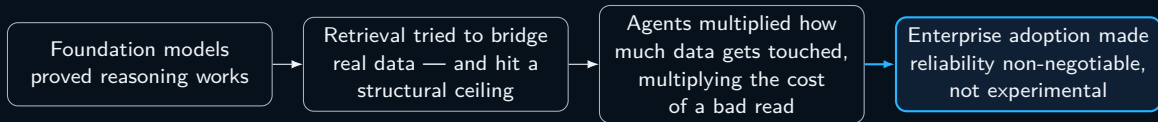
### With Citadel



Extraction cost paid once. A model swap touches only the substrate.

**Infrastructure persists for decades.**

# Why Now



**The missing layer was survivable when AI was experimental. It stopped being optional the moment reliability became the product.**

## The Foundational Layer Every AI Company Will Require

Any AI product reasoning over real documents faces the same dependency — and a competitor built on a solved version of it holds a compounding advantage.

### Not a Feature. A Survival Dependency.

- ▶ An extraction error is not local — every downstream layer inherits it.
- ▶ **The gap compounds, it does not reset.** Each model release, a Citadel-based competitor re-embeds in minutes; one without it re-runs extraction from scratch.

### Who Already Depends On It

- ▶ **New AI companies**, at the build-vs-buy decision for ingestion.
- ▶ Enterprises building internal AI over their own document corpora.

**Payphones did not lose to a better payphone. They lost when the infrastructure underneath calling changed.**

so how many tables did you say the excel file had and how many rows in each one?

From this workbook, I identified **3 logical tables** on the single worksheet.

| Table   | Approximate Rows | Number of Rows | Description                                                                      |
|---------|------------------|----------------|----------------------------------------------------------------------------------|
| Table 1 | 3–18             | 16 rows        | Black-Scholes option pricing model (lookup table, inputs, outputs, calculations) |
| Table 2 | 22–24            | 3 rows         | Dataset metadata (World Development Indicators source information)               |
| Table 3 | 25–62190         | 62,166 rows    | World Development Indicators dataset                                             |

Total

- **Worksheets:** 1
- **Logical tables:** 3
- **Total rows across all tables:** 62,185 rows

Notes

The third table is by far the largest:

- **Rows:** 62,166
- **Columns:** roughly 70 (Country Name, Country Code, Indicator Name, Indicator Code, followed by yearly columns from 1960 through 2025)

The first table is the trickiest because it's **not a clear rectangular dataset**. It's really a financial worksheet composed of several related blocks (normal distribution lookup, input parameters, calculated values, call/put Greeks, etc.). A naive person might incorrectly split it into 4-6 separate tables, while a semantic person would

ChatGPT ▾

 Share ...

Über additive Eigenschaften von Za...

PDF

what are these documents about

I can do that, but I don't currently have access to the contents of the documents in this chat.

Could you either:

- **attach the documents again**, or
- tell me **their filenames**?

Once I have them, I'll explain for each one:

- what the document is,
- what it's used for,
- what kind of data it contains,
- how it's structured (tables, sheets, sections, etc.),
- and any interesting or unusual characteristics.



## 2. Detailed Data Sections & Row Counts

### Sheet 1: `Sheet1`

`Sheet1` contains 5,785 total occupied rows spanning several distinct data tables/blocks:

- Option Distribution Calculations (Rows 3–5): 3 rows
- Option Inputs & Parameters (Rows 7–12): 6 rows
- Calculated Financial Values (Rows 14–18): 5 rows
- Data Source Metadata (Rows 19–23): 5 rows
- Main Dataset (World Development Indicators) (Rows 25–5785):
  - Header row: Row 25
  - Data rows: Rows 26–5784 (5,759 rows)
  - Total row: Row 5785 (*Total Result*)

### Sheet 2: `Pivot Table_Sheet1_1`

This sheet contains a single Pivot Table with 181 total rows:

- Header Row: Row 1 ( `Country` , `Year` , `Sum - Total` )
- Data Rows: Rows 2–180 (179 country summary rows)
- Summary Row: Row 181 ( `Total Result` : 11,784 )

👍 🗨️ ↻ 📄 ...

+ Ask Gemini

Flash ▾



[← B](#)

ASK

## Chat with this library

[📄 Export](#)[🗑️ Clear](#)

how many tables are in test(1) excel file and how many rows in each table?

The file `test(1).xlsx` contains 12 tables on Sheet1. The row counts for each table are as follows:

- Table #1 has 2 rows [test(1).xlsx (Sheet1) #1].
- Table #2 has 1 row [test(1).xlsx (Sheet1) #2].
- Table #3 has 4 rows [test(1).xlsx (Sheet1) #3].
- Table #4 has 5 rows [test(1).xlsx (Sheet1) #4].
- Table #5 has 5 rows [test(1).xlsx (Sheet1) #5].
- Table #6 has 1 row [test(1).xlsx (Sheet1) #6].
- Table #7 has 3,020 rows [test(1).xlsx (Sheet1) #7].
- Table #8 has 1,510 rows [test(1).xlsx (Sheet1) #8].
- Table #9 has 824 rows [test(1).xlsx (Sheet1) #9].
- Table #10 has 1 row [test(1).xlsx (Sheet1) #10].
- Table #11 has 179 rows [test(1).xlsx (Sheet1) #11].
- Table #12 has 180 rows [test(1).xlsx (Sheet1) #12].

what are these documents about? give me an overview

The provided documents cover a diverse range of topics including legal proceedings, academic credentials, technical architecture, mathematical theory, and international regulations.

**Legal and Regulatory Documents** Several documents detail legal frameworks and specific disputes. A High Court directive from Malaya outlines new classification codes for registering court cases effective January 1989 [135. Arahan Amalan Bil 5 Tahun 1988.pdf p1-p12]. Malaysian consumer protection laws are described in excerpts from the *Consumer Protection Act 1999*, which establish the Tribunal for Consumer Claims and define remedies for unsafe goods or services [CONSUMER PROTECTION ACT 1999 AMENDMENT 2019 .pdf p1-p114]. A specific consumer dispute is documented where Masum Billal seeks a refund and damages from ICONIX CO-LIVING SDN BHD regarding a tenancy agreement and access card issues in Kuala Lumpur [defence.pdf p1-p72]. Additionally, a Kazakhstani order establishes rules for issuing residence permits to foreigners, detailing required documents and processing timelines [98630\_rus\_20260706 (1).pdf p1-p58].

**Academic and Professional Credentials** The collection includes academic transcripts and certificates for Masum Billal. An official transcript lists 18 credentials earned between April 2025 and March 2026 from Coursera, Sophia Learning, and the Foundation for International Business Administration Accreditation (FIBAA) [academic\_transcript20260625-31-2kw0yq.pdf p1-p9]. A character certificate issued by the Sabujbagh Police Station in Dhaka verifies his antecedents [pcc-2025.pdf p1-p2]. Furthermore, a letter from the Ministry of Artificial Intelligence and Digital Development of Kazakhstan confirms his qualification as an 'Artificial Intelligence Engineer' [Digital nomad certificate.pdf p1-p4].

**Technical Architecture and Data Systems** Multiple documents describe data-intensive systems and architectures. *Designing Data-Intensive Applications* provides a technical overview of reliability, scalability, and distributed system patterns like consensus algorithms and stream processing [Martin-Kleppmann—Designing-Data-Intensive-Applications\_-O'Reilly-Media-(2017).pdf]. Another text, *Information Systems for Business and Beyond*, defines information systems components and discusses trends like the Internet of Things and blockchain [Information Systems for Business and Beyond-1.pdf]. The 'Citadel' platform is introduced as a stateful infrastructure layer that automates data extraction from unstructured files to enable compliance audits [citadel.md p1; Citadel Pitch Deck.pdf p1-p14]. Research on Pytheas, a system for discovering tables in CSV files, is also included [p2075-christodoulakis.pdf p1-p15].

**Mathematical Research** A significant portion of the evidence consists of excerpts from advanced mathematical treatises. *Über*

## A Harder Test, Not an Easier One

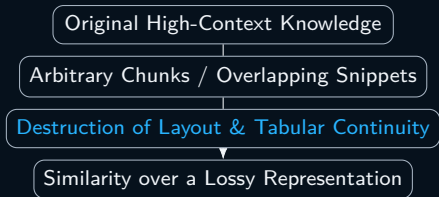
| Comparison                | ChatGPT              | Gemini               | Citadel         |
|---------------------------|----------------------|----------------------|-----------------|
| Files given               | 1 excel+1 800 pg     | 1+4 pdf              | 29 multimodal   |
| Tables                    | 3                    | 6                    | 12              |
| Rows (ground truth: 5733) | 60k+                 | 5785                 | 5733 (verified) |
| Searchable Corpus         | hard platform limits | hard platform limits | unlimited       |
| Table query               | Isolated to 1 excel  | Isolated             | All files       |

**Citadel solved the much harder problem at unprecedented scale including a single pdf of 800+ pages: searching the full corpus and still matched the verified ground truth.**

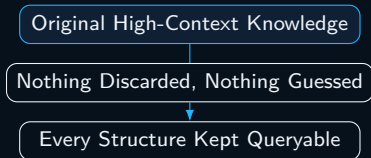
# Retrieval Cannot Recover What Structurally Disappeared

If a representation discards structure, nothing downstream recovers it. Every answer must trace back to exactly where it came from — nothing flattened, nothing untraceable.

## The Retrieval Failure Mode



## The Citadel Paradigm



## Enterprise First, Infrastructure Next

### Phase One: Direct to Enterprise

- ▶ **Legal** — multi-contract structural risk analysis.
- ▶ **Financial** — cross-document reconciliation audits.
- ▶ **Real Estate** — title, lease, and property-record intelligence.
- ▶ **Regulatory** compliance and due diligence.

### Phase Two: Infrastructure

Once proven directly with enterprise buyers, the same substrate opens as the ingestion and retrieval layer other AI companies build on — sold as platform, not a point solution.

**One layer proven with the enterprises, then opened to everyone.**

## Two Tiers, Priced on Usage

### Structure

Lossless ingestion into SQL-queryable tables. No retrieval, no Q&A — the raw substrate itself, billed on volume ingested and retained.

### Search

Everything in Structure, plus retrieval and natural-language Q&A over it — billed on volume plus queries executed.

Annual enterprise contracts in Phase One — metered API access once the substrate opens as infrastructure in Phase Two.

**Priced on what it does for you, not seats or licenses.**

## The Ask

Raising **\$3,000,000** for **10%** of the company (\$30M post-money) — priced for a working system, not a concept.

### What This Round Is For

- ▶ Founder, full-time.
- ▶ First 1-3 hires beyond the founder.
- ▶ Compute and infrastructure to deploy into real cloud infra and onboard real users.
- ▶ Legal, compliance, and IP setup for enterprise-grade deployments.

### What It Buys

- ▶ 18–24 months of runway.
- ▶ The technical barrier is already crossed — this capital converts a working system into paying design partners.
- ▶ Target: signed pilots, hardware/dedicated cloud, human and other resources, production engineering

**We already built it. We are raising to prove the market has demand for it.**

## Team

### Masum Billal

Founder

Solo-built: ingestion pipeline, query engine, and everything else this deck shows.  
Every result in this deck is measured against a working system, not a mockup.